

Features

- **Fast:** It is ten times faster than SQLite and delivers the best performance.
- **Object API:** It builds objects from the ground up, i.e., there are no ORM, no SQLite, no rows and no columns.
- **Powerful query builder:** Simple queries for objects with consistent checks at the time of compilation.
- **Cross-platform**
- **Simple threading:** Objects returned by ObjectBox work in all threads with no strings attached.
- **DaoCompat library:** This is an add-on library for greenDAO APIs for ObjectBox.
- **Support:** It has an active community with more than 150,000 active users, and a strong team for active beta, unit and final testing.

Official website: <http://objectbox.io>

Latest version: 1.4.1

Realm DB

Realm DB (Real Mobile Database) is an open source database specifically designed for mobile devices from scratch, and is available for various platforms including Xamarin and React Native.

It is serverless and cross-platform like SQLite. Data can be stored on the disk as well as in memory.

The key difference between Realm and ORMs is that the former is not an abstraction built on top of SQLite, but a whole new database engine. Rather than a relational model, it is based on an object store. Its core consists of a self-contained C++ library. It currently supports Android, iOS (Objective-C and Swift), Xamarin and React Native.

Realm is under active development and has been releasing updates pretty often. The development team is very responsive on the database's issue tracker and StackOverflow.

Features

- **Easy-to-use:** Realm is very easy to use compared to CoreData and SQLite. Realm data models are defined using traditional NSObject-style classes with @properties. You can simply subclass RLMObject to create your Realm data model objects.
- **Faster response:** Realm is faster than other databases like SQLite, FMDB, CoreData and Couchbase Lite in app data storage. Even complex queries take nano seconds, and stay up-to-date with new data.
- **Realm browser:** This allows users to explore and administer realm databases.
- **Cross-platform:** Available for Android, iOS and other popular mobile operating system platforms.
- **Data security:** Provides enhanced security of data via transparent encryption and decryption.
- **Concurrency:** Realm handles concurrency with MVCC architecture, and these views 'automatically update' when a transaction is committed from any thread.

- **Support:** Realm is highly supported via proper documentation, the community and is always updating to the latest features. It supports Objective-C as well as Swift, which suggests that this could be a long-term solution for iOS developers.

Official website: <https://realm.io/>

Latest version: 4.3.1

Couchbase Lite

Couchbase Lite is a fully-featured open source NoSQL embedded JSON database designed for mobile devices. Data in Couchbase Lite is stored as JSON documents. Each document may have one or more attachments, which are essentially uninterpreted binary data that is stored and loaded separately from the document itself. Couchbase Lite supports persistent indices (called views), and uses MapReduce to manage and query them.

Couchbase Mobile is the solution provided by Couchbase Lite for mobile applications. It comprises three different components: Couchbase Lite, an embedded NoSQL database and Sync Gateway. Couchbase is an offline-first database, and syncs with the cloud when needed or when the network is available. Couchbase Lite runs locally on the device and persists data as JSON and binary format. All CRUD operations are performed on the local database. Developers do not need to write sync code (if needed) to sync the local database with the cloud, as this is handled by Sync Gateway.

Couchbase Lite provides native APIs for Android and iOS with plugins for Xamarin and PhoneGap.

Features

- It is lightweight — just 500KB to 1MB.
- Data is secured on the mobile device using 256-bit AES encryption.
- Synchronisation of data is from device-to-device via a peer-to-peer network.
- Contains both REST APIs and native APIs for data management.
- Cross-platform support available for Android, Mac, Windows and Linux.

Official website: <https://www.couchbase.com/products/lite>

Latest version: 5.0

LevelDB

LevelDB is an open source database for mobile and Web apps, and was designed and developed by Jeffrey Dean and Sanjay Ghemawat of Google Inc. It is a simple, key-value store database built by Google. It is currently used in Google Chrome and other software. It supports arbitrary byte arrays as both values and keys; singular *put*, *get* and *delete* operations; batched *put* and *delete*; and bi-directional iterators. Simple compression is performed using the fast Snappy algorithm.